

AGRICULTURAL SUPPLY CHAIN CASE STUDY

Understanding the gap between farmer earnings and consumer prices.

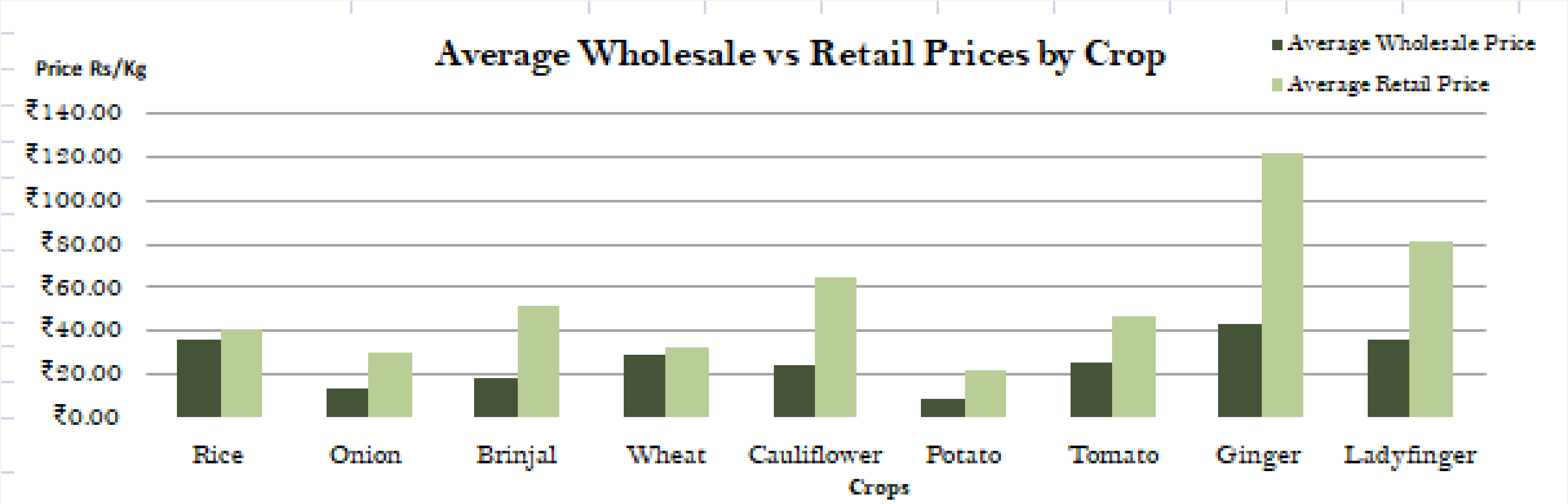
Kamakshi Dhamija
Data Analytics

THE PRICE GAP PROBLEM

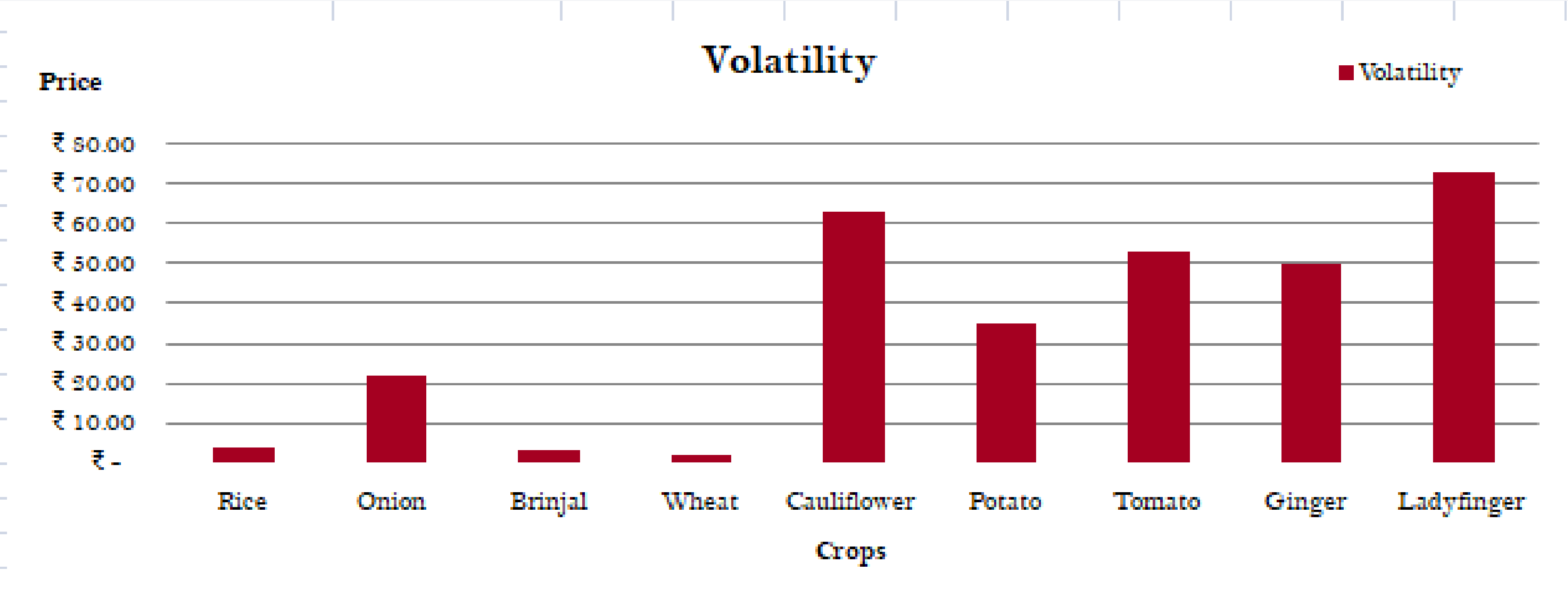
FARMERS OFTEN RECEIVE LESS THAN HALF OF THE FINAL CONSUMER PRICE.

Crop	Overall Avg Farmer Share %
Rice	88.42%
Onion	48.81%
Brinjal	35.08%
Wheat	88.61%
Cauliflower	40.32%
Potato	37.15%
Tomato	54.38%
Ginger	36.75%
Ladyfinger	42.10%

DATA ANALYTICS DASHBOARD



VOLATILITY AMONG CROPS



WHAT DRIVES HIGH VEGETABLE PRICES?

Variable Pair	Correlation
Shelf Life vs Markup %	-0.498
Transport Burden vs Markup %	0.356

- Shorter shelf-life crops showed higher markups, indicating spoilage risk strongly affects retail pricing.
- Crops with higher transport burden also recorded higher markups due to logistics and distribution costs.

KPI ANALYSIS

KEY PERFORMANCE INDICATORS	
Highest markup crop	Ginger
Lowest markup crop	Wheat
Highest seasonal markup	Kharif
Average market markup	131.95%
Largest price gap	Ginger (78.92)

- Ginger recorded the highest markup and largest price gap, showing significant retail inflation for perishable crops.
- Wheat maintained the lowest markup percentages, indicating relatively stable pricing and lower supply chain costs.
- Kharif season showed the highest overall markups, while the average market markup exceeded 131%, highlighting substantial gaps between wholesale and retail prices.

SO WHAT NEEDS TO CHANGE?

Improve Cold Storage Infrastructure

Perishable crops showed the highest markups, highlighting the need for better cold-chain and storage facilities.

Reduce Transport Inefficiencies

High transport burden was linked with higher retail prices due to spoilage and logistics costs.

Increase Price Transparency

Real-time mandi price systems can help reduce information gaps between farmers, traders, and consumers.

Support Direct Farmer Procurement

Reducing intermediary layers can improve farmer earnings while lowering consumer prices.

Built using Excel, Agmarknet & FSD Delhi market data.